



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRE AMMAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

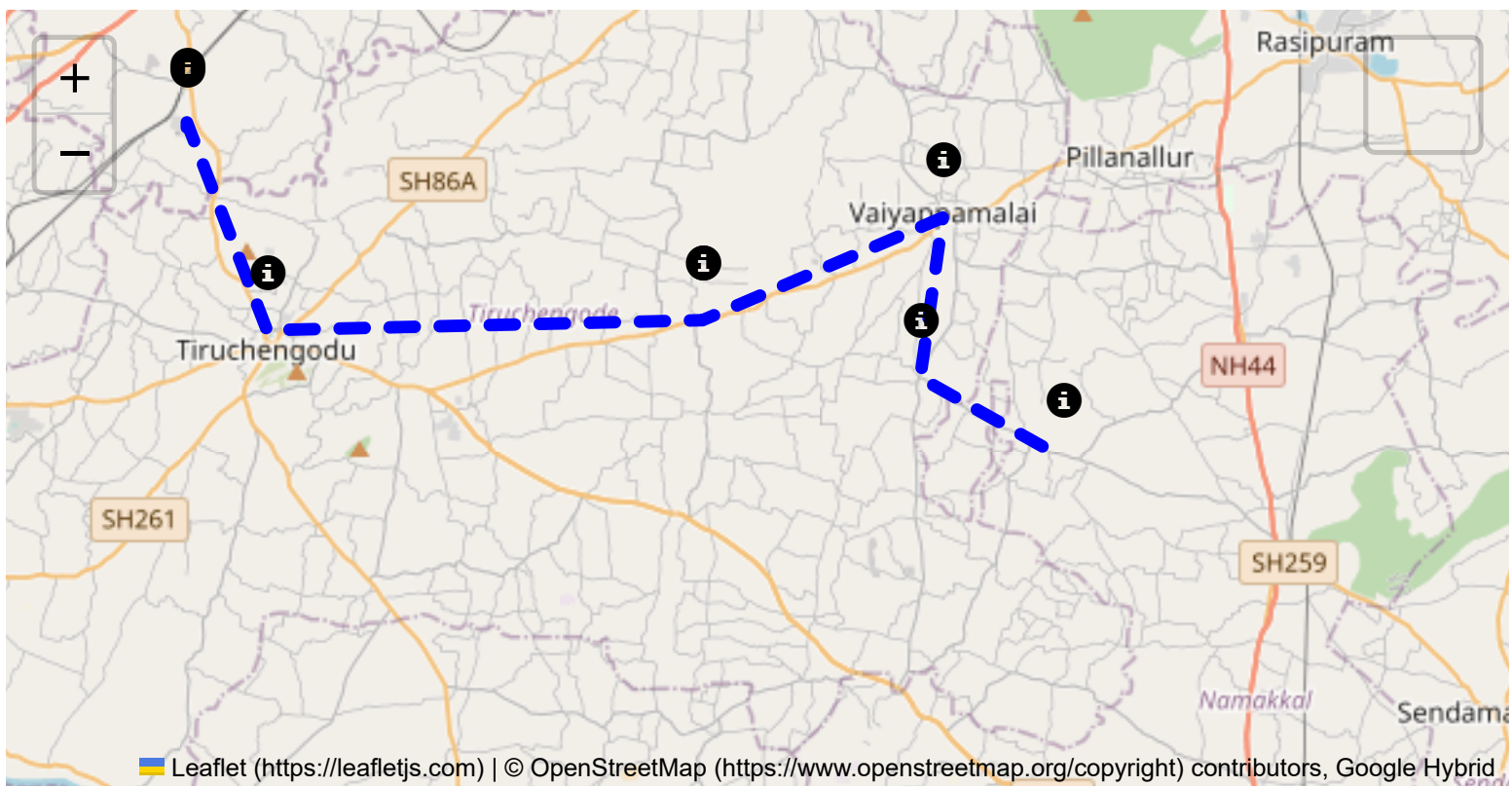
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

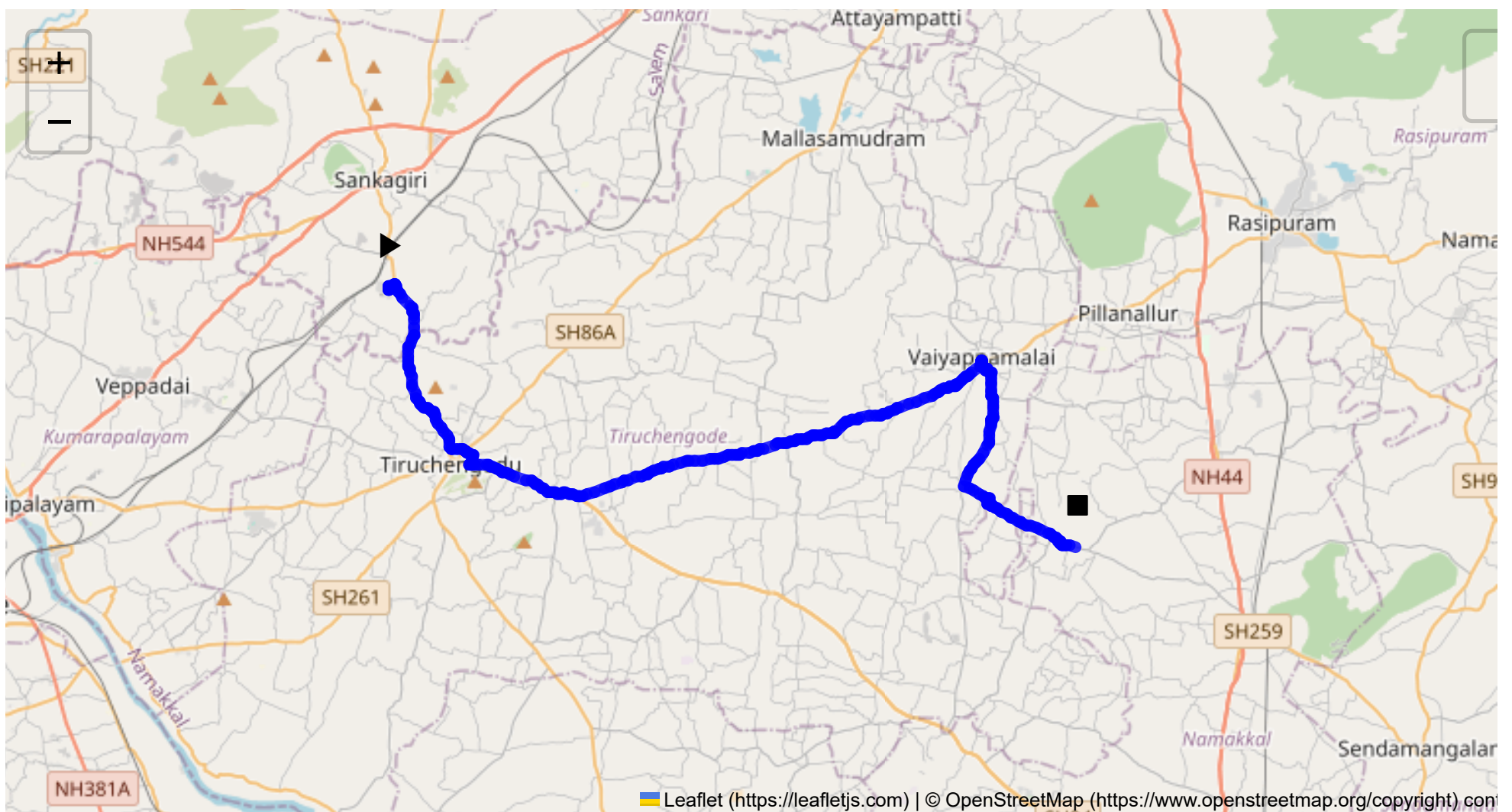
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 40.36 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.349681, 78.11367)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Elur via Puthur, Tiruchengode, Elachipalayam, Vaiyappamalai, and Periamanali covers approximately 40.36 kilometers. The route primarily follows regional roads and State Highways, characterized by intersections in towns and some rural segments, which include various turns and require navigation through semi-urban and rural areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu typically experiences a tropical climate with heavy monsoons from October to December and temperatures ranging from mild in winter to very hot in summer. The risks include heavy rain during the monsoon leading to slippery roads and potential flooding in low-lying areas, affecting traction and visibility. During the dry season, dust and glare from the sun could also hinder visibility.

3. Analysis of Traffic Patterns:

- **Peak Hours:** Morning (7-10 AM) and evening (5-8 PM) see increased traffic due to local commuter movements.
- **Congestion-Prone Areas:** Tiruchengode around the central bus stand due to public transport and local shops, and Puthur crossroads where local traffic converges.

4. Assessment of Road Quality and Infrastructure:

- **Road Quality:** Generally, mixed; main highways like SH 79 are in better condition compared to smaller roads which might have potholes and uneven surfaces.
- **Infrastructure:** Adequate in larger towns with signals and signage. However, rural stretches may lack road markings and lighting, making night driving riskier.

5. Suggestions for Alternative Routes for Emergencies: Consider bypassing congested towns and using SH 86, directly connecting Sangagiri and Vaiyappamalai, which might be beneficial during emergencies or road closures.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Adherence to the Indian Motor Vehicle Act regarding the transport of hazardous materials is essential. Specific routes and timings may be restricted for such transports to avoid densely populated areas during peak times.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Specific data on incidents may be sparse; however, typical issues involve accidental spills, particularly in areas with sharp turns or poor road conditions. Past incidents often occur at intersections or due to mechanical failures on steep segments.

8. Environmental Considerations and Sensitive Areas: There are agricultural fields along the route, with potential ecological impact in case of a hazardous material spill. Protecting water bodies and farmland from contamination is crucial.

9. Analysis of Communication Coverage, Noting Potential Dead Zones: Coverage is generally good in towns but may be patchy in certain rural stretches, particularly between settlements. It's advisable to have reliable communication devices with emergency frequencies preset.

10. Estimated Emergency Response Times for Different Route Segments:

- **Sangagiri to Tiruchengode:** 20-30 minutes, moderate accessibility.
- **Tiruchengode to Vaiyappamalai:** 30-40 minutes, depending on congestion.
- **Vaiyappamalai to Elur:** 30 minutes, with quicker response in peri-urban areas.

12. Overall Summary of Risk Assessment: The route involves moderate risks primarily tied to weather, road quality, and traffic congestion. With careful planning, adherence to regulations, and awareness of

local conditions, risks can be mitigated. Drivers should maintain caution during adverse weather, be mindful of road conditions, particularly in rural areas, and prepare for longer emergency response times in remote sections. Regular communication and route updates will enhance safety during transportation of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	High	11.38190, 77.90254	15 KM/Hr
7	Turn	Medium	11.38182, 77.90259	30 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.41410, 78.08065	10 KM/Hr
10	Turn	Medium	11.41321, 78.08067	30 KM/Hr
11	Turn	Medium	11.41276, 78.08042	30 KM/Hr
12	Turn	High	11.41113, 78.08228	15 KM/Hr
13	Turn	High	11.41077, 78.08224	15 KM/Hr
14	Turn	High	11.40979, 78.08384	15 KM/Hr
15	Turn	High	11.39425, 78.08455	15 KM/Hr
16	Turn	Medium	11.39350, 78.08368	30 KM/Hr
17	Turn	High	11.37119, 78.07401	15 KM/Hr
18	Turn	High	11.36691, 78.08320	15 KM/Hr
19	Turn	Medium	11.36684, 78.08323	30 KM/Hr
20	Turn	High	11.36525, 78.08282	15 KM/Hr
21	Turn	High	11.36462, 78.08383	15 KM/Hr
22	Turn	High	11.36435, 78.08389	15 KM/Hr
23	Turn	Medium	11.35168, 78.10819	30 KM/Hr
24	Turn	High	11.35121, 78.10818	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41410, 78.08065



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41321, 78.08067



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41276, 78.08042

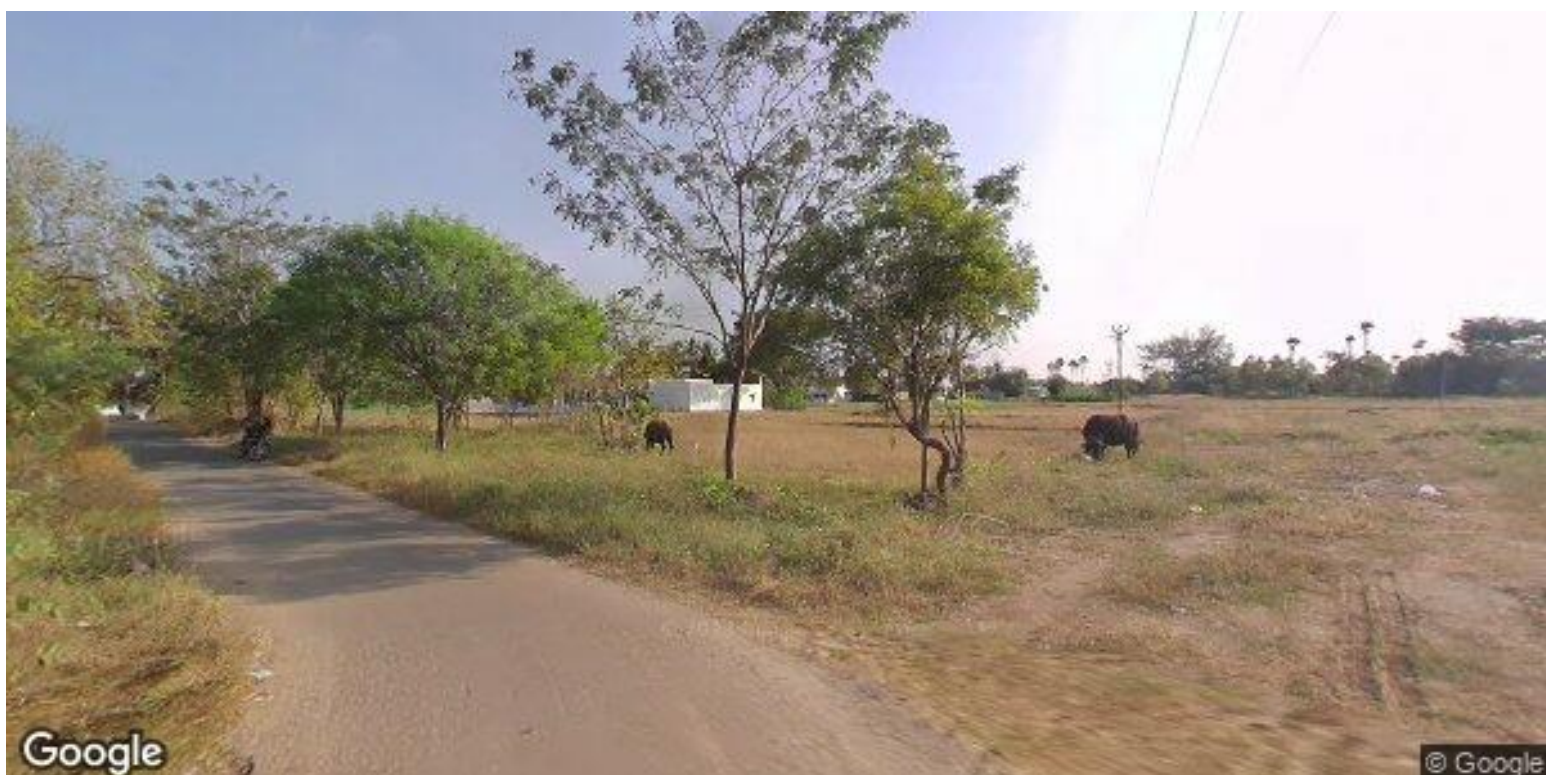


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41113, 78.08228



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41077, 78.08224



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40979, 78.08384



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39425, 78.08455



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39350, 78.08368



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37119, 78.07401



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36691, 78.08320



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36684, 78.08323



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36525, 78.08282



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36462, 78.08383



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36435, 78.08389



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.35168, 78.10819



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.35121, 78.10818
